

**Seventh Semester B. Tech. (Electronics and Communication
Engineering) Examination**

EMBEDDED SYSTEMS

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions carry marks as indicated.
 - (2) Assume suitable data wherever necessary.
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1. Illustrate the booting process in Cortex M microcontroller and explain what happens when the external reset is pressed with a neat flow chart. 6(CO1)
 2. Interpret what do you understand by volatile variable ? Explain where and why it is used with the help of an example. 6(CO2)
 3. Interpret Race condition and its side effects in embedded systems. 6(CO3)
 4. Clarify the use of a header file and a class specified in header file. Demonstrate the use of class with a sample program to control a GPIO pin in microcontroller ? 6(CO3)
 5. Detect how floating point to fixed point conversion is carried out considering Q4.3 and solve for value "-3.97". 6(CO3)
 6. Demonstrate a decimal up/down counter on a interfaced seven segment display (common anode) at PORT A and a single switch (active low) at PORT B. The switch connected at port B acts as an interrupt that controls the change in direction of counting. Provide all calculations of hex table, with Embedded C Code to demonstrate the counter functionality. 10(CO4,5)

7. Check the implementation of a pulse width modulation with Cortex M microcontroller for the DC motor interfaced at GPIO pin. 10(CO4,5)
8. Justify the need of SPI protocol to illustrate how 8x8 dotmatrix display (MAX7219) can be interfaced for displaying a character of your choice. All calculations, display character and multiplexing requirements should be discussed in your answer in detail in regards to SPI protocol. 10(CO4,5)

